BINNARY CONTOUR IMAGE DOWN SCALING FOR PIXEL ART

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ABSTRACT

Pixel art is an old technique that has been used to create graphics for video games. It is an effective means of creating row resolution displays. It takes designers a long time to draw pixel art on a pixel-by-pixel basis. If it were possible to create pixel art more easily, the field of pixel art could be expanded as a method of expression.

We first investigated the procedures that are used for drawing pixel art and the techniques for connecting the direction of contour lines in order to understand all of its features. The required width of contour lines used in pixel art is often just one pixel. However, existing methods are unable to resolve single pixel contour lines. It is also difficult to draw contour lines for small images. We have to scale down binary contour lines to create an image using existing methods, or else they become dull grayscale images or lose their information. To solve this problem, we propose a method of creating contour line images by taking photographs and scaling them down. We can then draw binary contour line images for pixel art.

1. INTRODUCTION

Pixel art was commonly used to create graphics for old video games. This is an effective means of creating row resolution displays. It takes games illustrators a long time to draw pixel art on a pixel-by-pixel basis. Video game publishers often decide to use polygons and illustrations to make video games, because they want to reduce production time and improve the performance of computers. But pixel art is sometimes more impressive than polygons. In addition, a toy[1] has been released. The toy has pins and panel. Users put pins to panel, then can make pixel art. If it were possible to create pixel art more easily, the field of pixel art could expand as a method of expression. The purpose of our research is to find a way of drawing contour lines of the size required for pixel art from photographs.

2. ANALYSIS OF DRAWING USING PIXEL ART

2.1. Features of Pixel Art

We will now explain the main features affecting the use of pixel art.

1. Used less than QVGA
   Pixel art is used less than QVGA displays for video games. Polygons and photographs/illustrations are more frequently used, because more than VGA displays have become more common.
2. Sprites
   Sprites are used to animate without redrawing the whole display. This technique overdraws arbitrary parts of the image (in many cases, 16×16 px). It was convenient for low spec computers to be able to animate without redrawing the whole display.
3. Creation of big pictures by repeating textures
   For example, repeating the image of a tree to express a forest.
4. Limitation of colors (4, 16, 32, 64)
   Sprites utilize a color palette, and these palettes contain a limited number of colors. The limitations are 4, 16, 32, and 64 colors, etc. Moreover, the number of characters can be increased by replacing the palettes used for the sprites.
5. Emphasis (color, shape, animation)
   The pixel counts of images are limited, e.g.16×16 px or 64×64 px. As far as possible, pixels are used and blank spaces are minimized. Shapes are often deformed. The emphasis is on color, shape, and animation. Contour lines are often drawn to give emphasis.
6. No alpha channel
   "No alpha channel" means that there is no alpha channel on the palette. An image can appear as a transparent color. The availability of transparent colors depends on the computer that is used, for example, the first color of the palette or the first pixel of the image. Creators often drew pixel art by adjusting to these specifications. If creators want a translucent effect, the game program must render the translucent image uniformly.
7. Parallel projection, fixed view point
   When a 3D shape is expressed in pixel art, parallel projection is often used to fix the aspect ratio. The purpose of this is to reuse data and to give emphasis.

2.2. Procedure for Drawing Pixel Art

According to Suguru.T[2], the procedure for drawing pixel art is roughly as follows. However, procedures 6 and 7 might not be carried out.

1. Preparing the reference image
   It is difficult for a human to draw a picture without looking at a reference image.

2. Drawing pixel art into a few pixels
   A pixel count of the 'canvas' is often not large enough to express of all of the desired elements. Creators must choose which elements to draw. Contour lines are often drawn in black.

3. Thinning of lines
   The investigation in 2.3 indicates that the width of contour lines is often just 1px. Moreover, if two lines are fused, then one line is often omitted.

4. Rough coloring
   Rough coloring helps creators to visualize the final drawing.

5. Shading
   Creators draw in shade and highlights. The image begins to look solid.

6. Coloring of contours
   If this process is not carried out, the contour lines are often left as black. In this case, the black-contoured pixel art will make a deeper impression than other pixel art. Many pixel art images are not used alone, and are often arranged to overlap with other pixel art images.

7. Anti-aliasing
   Anti-aliasing means that images are given only internal contour lines in many cases, because the external lines are piled onto other unspecified images that use transparent colors. There are no anti-aliasing techniques that can be matched to all conditions.

The pixel art is usually hand-drawn when these procedures are employed.

2.3. Survey of Pixel Art Contour Lines

To clarify the methods used for drawing contour lines, we investigated about 200 examples of pixel art. The pixel art that we investigated range from 15×15 px to 152×102 px, and are not repeated images. Moreover, the color of the edges was drawn in black to delineate the contour uniquely. We investigated the ratio of contour line in the whole image, in addition to connecting edges.

2.3.1. Connection of Contour Lines

Fig.1 shows the result of our investigation into the connection of contour lines. This pattern is drawn on a contour line with an 8-neighborhood of 3×3 px. The numbers under each picture give the frequency of the appearance of each pixel. The top 100 among all of the 256 patterns that we examined are shown. From 1st to 12th place form a straight line of 1px in width for the 8-connection. From 13th to 20th form a branching straight line of 1px in width. This shows that 1px width lines are often drawn.

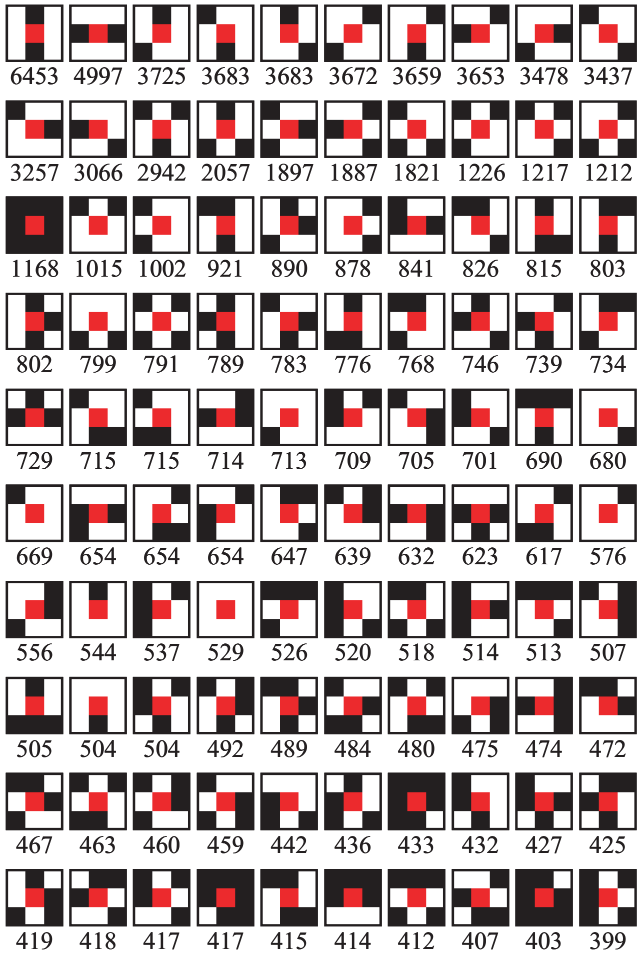

Fig. 1. Ranking of connections for pixel art contour lines

2.3.2. Ratio of Contour Line Pixels for the Whole Image

Fig.2 shows the ratio of the edge pixels for the whole image. On average, the straight lines appear at a ratio of 20% of the whole image, demonstrating that about 20% of the whole image is often used as a contour line.

Therefore, the assumptions made about the contours lines are that;

1. The contour lines must be formed by lines 1px in width in this investigation.

2. The ratio of the edge pixels must be about 20% of the whole image.

We propose that the algorithm for drawing contour lines is adapted in light of these two requirements.

2.4. Previous Work

Sometimes pixel art does not include contour lines, but mostly it does. In most cases the contour lines are drawn in color

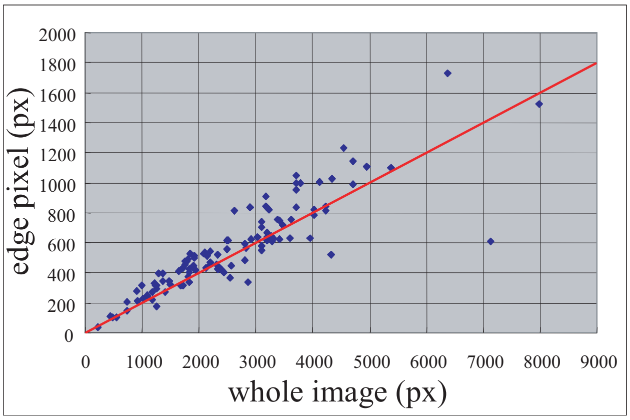

Fig. 2. Number of contour pixels used in pixel art

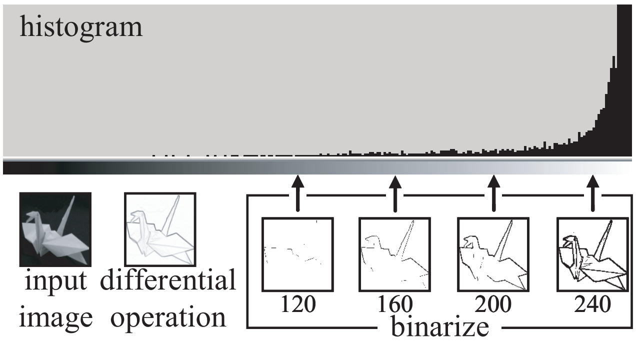

Fig. 3. Binarizing by thresholds

and are lines of 1px in width. If we binarize an image after applying a differential filter, grayscale contour lines can be obtained by the p-tile method or the mode method [3]. The mode method requires a histogram with a double peak, but differential histograms only have a single peak.

Pixel art now uses binary images and 1px-width lines. If we force a differential filter image to binarize (Fig.3), then the contour lines are not 1px in width. To achieve a 1px line-width, we must choose a threshold value: at a value of 120, lines are not formed. To get lines to form, we must choose a threshold of 240, in which case the lines are about 4px in width. To get 1px width contour lines to form, we should choose a threshold of 240 and apply thinning to it. This is the preferred connection for achieving beautiful lines in pixel art.

The overall pixel count in most pixel art is low. It may be easier to draw contour lines from a larger image than from a smaller image and then reduce the contour lines. We can use bi-linear interpolation, bi-cubic interpolation and the nearest neighbor method as resampling algorithms.

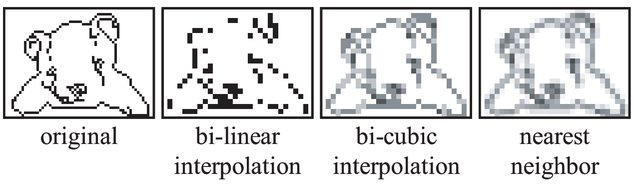

Fig. 4. Scaling down a binary image by existing methods

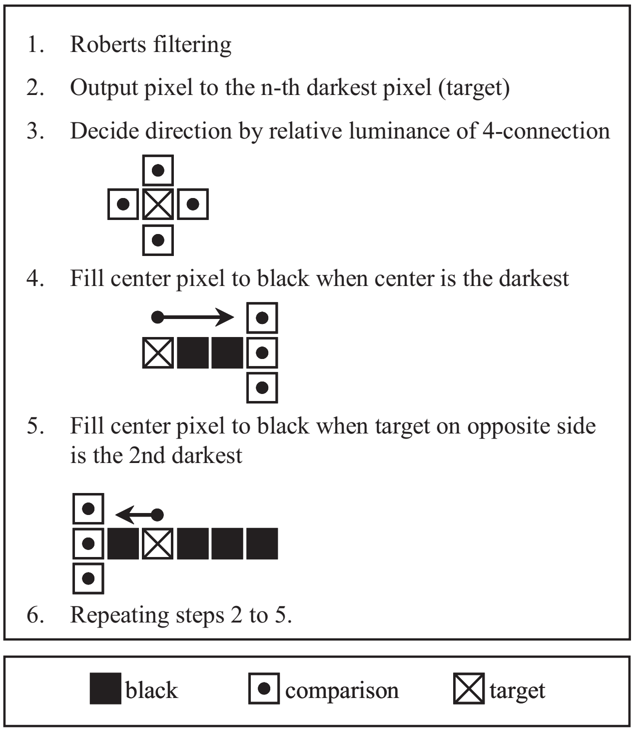

Fig. 5. Rules for drawing contour lines 1

This figure is reduced to half by each algorithm (Fig.4).

The number of colors is increased by using bi-linear and bi-cubic interpolations, so we then we cannot use binary image reduction. The nearest neighbor method maintains the binary image, but it does not include the contour lines. To solve these problems, we have proposed a method for drawing in contour lines of 1px width.

3. METHOD FOR DRAWING CONTOUR LINES

3.1. Summary of Proposed Method

The proposed algorithm has roughly 4 steps.

1. Draw the contour lines
   The contour line image is drawn with reference to the gray level by differential filtering.
2. Noise reduction
3. Contour line reduction
4. Repeat procedures 2 to 3

3.2. Drawing the Contour Lines

This method can be used to draw contour lines of 1px width using a algorithm that enables anti-aliasing at the same time. This anti-aliasing can be invalidated by setting it to the same color as the background.

Figs5 & 6 show the proposed algorithm.

Rule 1 is applied first, and if it matches the requirement of rule 2, then we proceed to rule 2. Fig.7 shows a

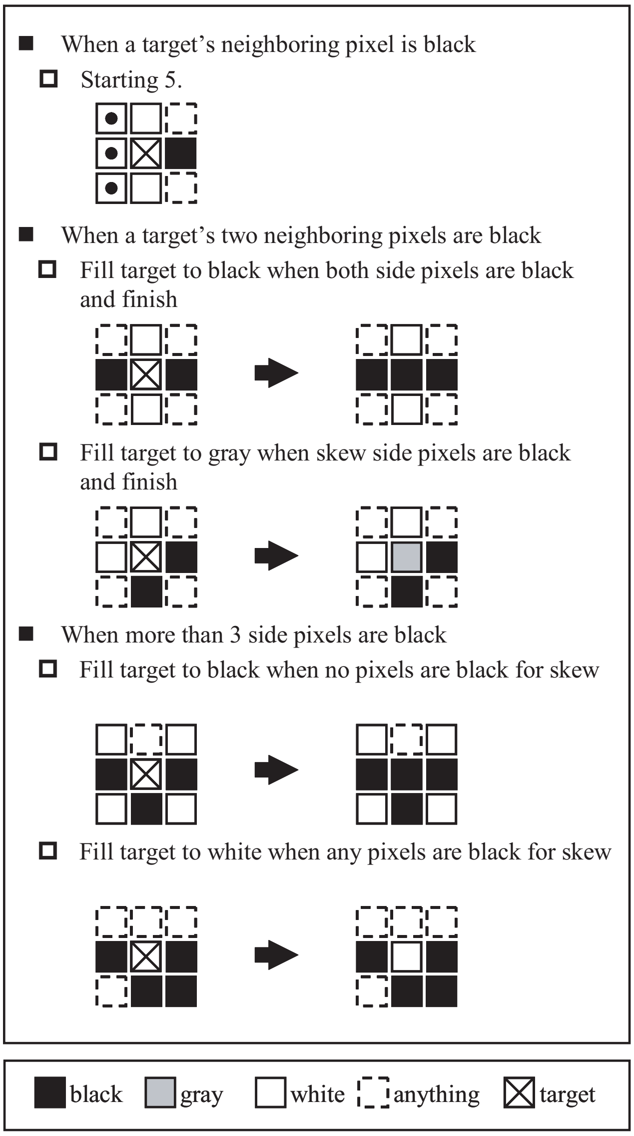

Fig. 6. Rules for drawing contour lines 2

straight line for which anti-aliasing is carried out by using this method.

3.3. Noise Reduction

The noise removal process is shown in Fig.8. We will call the first scenario in Fig.8 white noise, the second scenario is burr noise, and the third scenario is ladder noise.

A differential filter picks up the noise. The points are isolated by the 8-connection methodology. This connection ranks 64th in popularity according to the investigation in 2.3. This order is low, and so a point is deleted.

The definition of burr noise is that only 1px is adjacent to the straight line. Burr noise of this shape does not rank 100th in 2.3. Moreover, when anti-aliasing to burr noise exists in both neighboring cells, they are deleted because it becomes non anti-aliasing.

Ladder noise is a variety of burr noise. It appears as two scalar lines for similar reasons to burr noise.

Fig. 7. Example of drawing a contour line by the proposed method

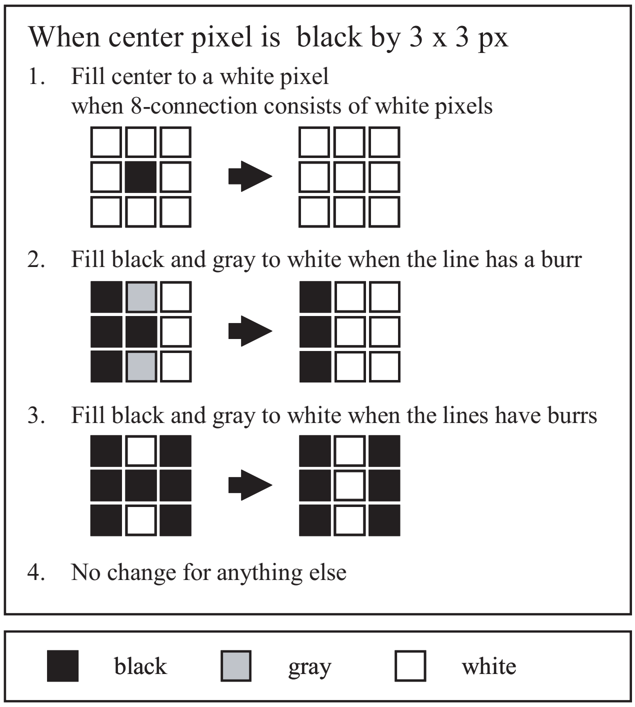

Fig. 8. Noise reduction

3.4. Binary Image Scaling Down

The method of scaling down has 4 steps.

Step1. Labeling of contour lines

Designers simplify regions on a part-by-part basis when they draw pixel art. However, it is difficult to make the computer distinguish the different parts. Instead, the contour lines are labeled according to the 8-connection and then the contour lines are subdivided to each layer (Fig.9). The layers are sorted according to the number of black pixels. The contour lines are made wider by each layer.

Step2. Widening contour lines on all layers

We now describe the making of the line-widening algorithm (Fig.10). We proposed the application of a contour-tracking algorithm. Now we decide that a mass of 4px is 1block. The algorithm searches for blocks in the order shown in Fig.10. It first searches for an output of 3px in a block, then of 2px, and finally of 1px.

If it finds a preferred pixel, then it fills the block to black. Because a straight line of 1px width is desirable, the order of the search in the figure takes this into consideration, according to the investigation in 2.3. In Fig.10, the left picture is filled to block.2,

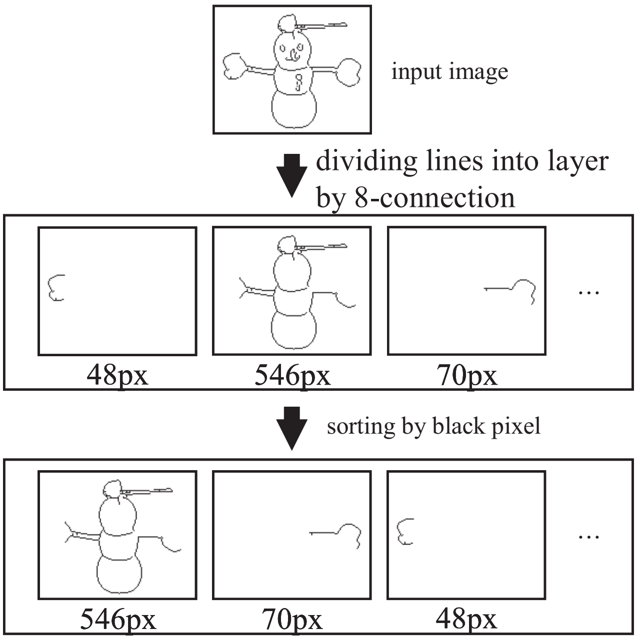

Fig. 9. Dividing a picture into layers

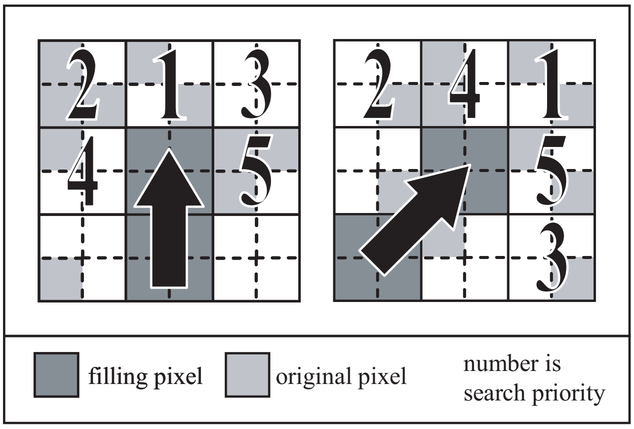

Fig. 10. Direction of search for widening a line

and right picture is applied to block.1 as black pixels. Fig.11 shows an easy execution example. The starting position for widening the lines is the longest straight line of the contour.

Step3. Integration of all layers
After all of the lines in a layer have been widened, the image is subdivided into the integrated layers. When the subdivided images are integrated, the integration methods produce 4 patterns (Fig.12). The method for selecting one from these patterns is as follows.

The first layer is used as a base, and the next or subsequent layers move.

Fig.13 shows an example in which the degree of freedom for integration is 2.

The upper images in Fig.13 show blocks that do not fit. They must move by 1px to the x-axis and to the y-axis. There are 4 patterns. Pattern a shows that

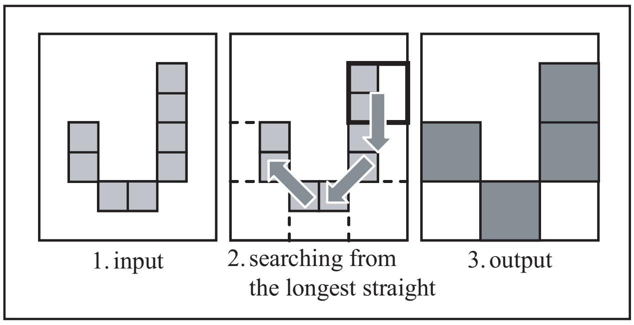

Fig. 11. Example of widening a line

the first layer and the second layer overlap by 1block, and thus the entropy is reduced. Pattern b shows an overlap of 1 block between the first layer and the second layer, so again, the entropy is reduced. Pattern c shows 4 blocks that make a bigger block and this is not the preferred form for pixel art. Pattern d is therefore the most promising of these patterns.

Step4. Simple scaling down
Scaling down of the widened lines by using a simple scaling-down algorithm can then be used to create a smaller image.

4. EXAMPLE

An example of an image created by the proposed method is shown in Fig.14. The input image pixel count is 150×122 px. This does not include the bounding box, and has little margin. Fig.14a becomes Fig.14b by Roberts filtering. Fig.14b becomes Fig.14c by the proposed method of drawing contour lines. Fig.14c becomes Fig.14d by the proposed method of noise reduction and invalidates anti-aliasing. Fig.14d becomes Fig.14e by the proposed method for the reduction of contour lines (first iteration). Fig.14e becomes Fig.14f by the proposed method for contour line reduction (second iteration). This is drawn with simple lines for arms. This is a good form for pixel art.

Fig.14e is doubled in size and Fig.14f is quadrupled.

Fig.15 shows an example of the proposed method. The numbers under pictures are the number of pixels in the bounding box. The bear shown in Fig.15 is a simple form of pixel art. But, contour lines may not be drawn in the case of the puppet because the lines are too close together.

5. CONCLUSION

The purpose of this research is to propose a method of drawing contour lines in pixel art, and especially small pictures of less than 100×100 px, by using a photograph. This paper shows the features of pixel art and describes our proposal for drawing pixel art contour lines.

Our method of drawing contour lines cannot ensure that the lines will connect. Therefore it cannot be used for colored regions when using paint software.

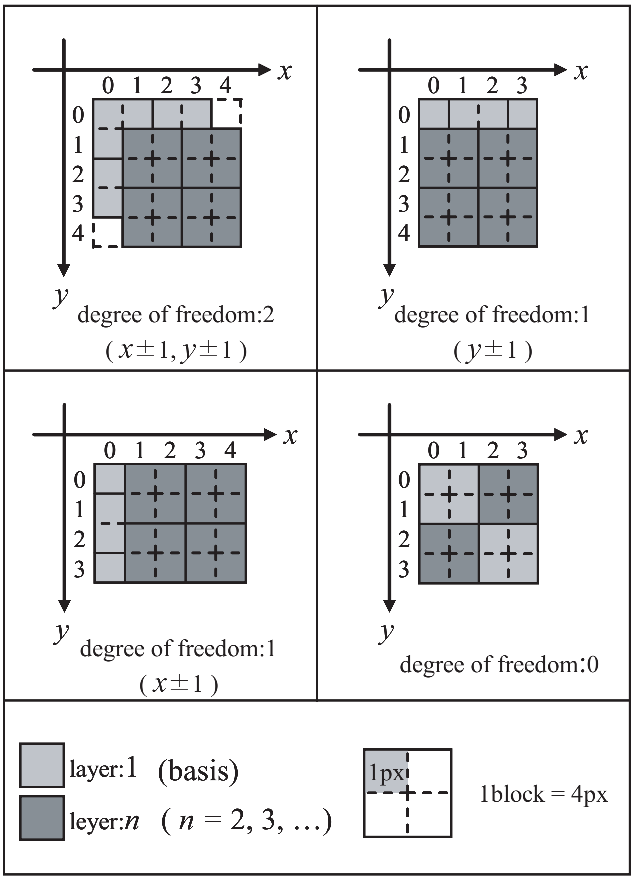

Fig. 12. Degrees of freedom of layers

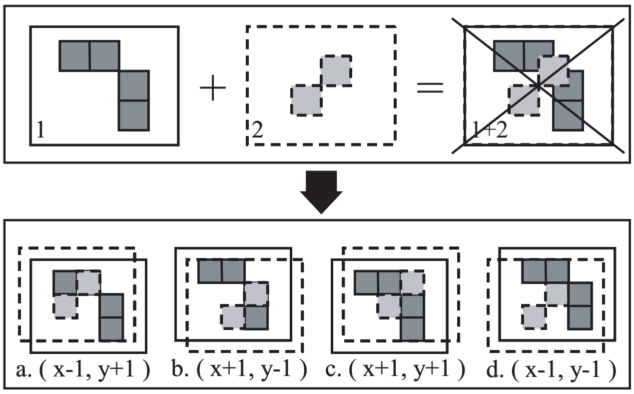

Fig. 13. Methods for integration of layers

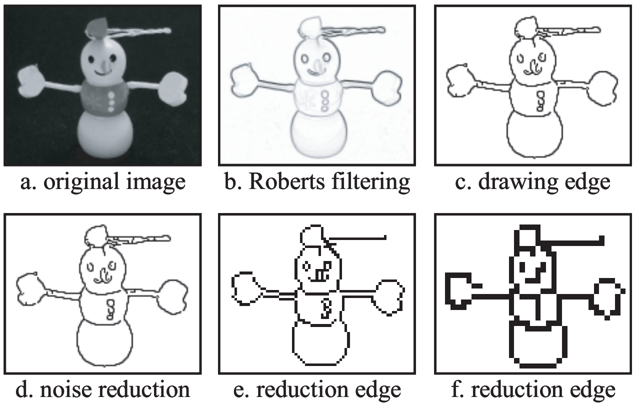

Fig. 14. Example of proposed method

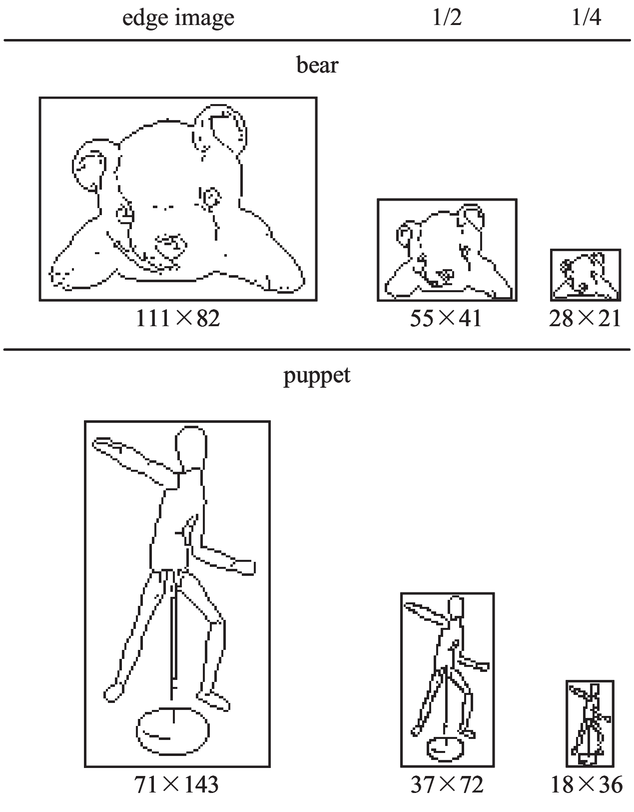

Fig. 15. Example of reduction

The scaling down algorithm can only reduce by a factor of $1/digital$ number. If there is a requirement for a size other than $1/digital$ number, one must first change to an integral size multiple and then apply our method.

Our next goals are to implement the use of color and to achieve suitable profiles by using geometrical data in pixel art.

6. REFERENCES

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